EILEEN SANTOS

Senior Data Engineer | Enterprise Data Architect | Cloud & Big Data Specialist

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PROFESSIONAL SUMMARY

Results-driven Enterprise Data Architect and Senior Data Engineer with 10+ years of experience designing, optimizing, and automating large-scale data pipelines and cloud-native architectures. Expert in ETL development, distributed computing, and enterprise data optimization using AWS (S3, Glue, Redshift, Lambda), Azure, GCP, Apache Spark, Databricks, and Kubernetes. Skilled at Infrastructure-as-Code with Terraform and CI/CD automation to deliver high-performance, secure, and compliant data ecosystems, translating complex engineering initiatives into measurable business outcomes in financial and enterprise environments.

Specializes in diagnosing and resolving performance bottlenecks in Amazon Redshift (query plan analysis, WLM tuning, compression, distribution/sort key redesign) and in Databricks Lakehouse architecture, Delta Lake optimization, and Unity Catalog governance across multi-cloud environments.

CORE COMPETENCIES

- **ETL & Data Pipelines:** Architected real-time and batch ETL workflows with Apache Spark, Airflow, PySpark, and SQL, improving efficiency by 40%.
- **Cloud Data Engineering:** Serverless data lakes and warehouses on AWS (S3, Glue, Redshift), GCP (BigQuery, Dataproc), and Azure (ADF, Synapse).
- **Streaming & Real-Time:** Kafka, Spark Streaming, and AWS Lambda for low-latency, event-driven architectures; reduced ingestion delays by 30%.
- **Credit Risk Decisioning:** Delivered PCI (Proactive Credit Line Increase) and HRAM (High-Risk Account Management) pipelines at Wells Fargo for automated risk segmentation and faster decisions.
- **Redshift Performance Tuning:** WLM, EXPLAIN plans, DIST/SORT keys, VACUUM/ANALYZE, compression encoding.
- **Databricks & Unity Catalog:** Centralized governance, lineage, access control, and data-quality enforcement across multi-workspace environments.
- **Microsoft Fabric & Azure:** Lakehouse, Pipelines, Semantic Models, Notebooks, Data Agents, and DirectLake for high-speed Power BI reporting; Synapse, Data Factory, Data Lake Gen2.
- **Data Governance & Security:** Improved data accuracy by 20% while ensuring GDPR, CCPA, SOX, and SOC2 compliance.
- **ML & AI Pipelines:** Scalable feature stores and ML-ready datasets supporting predictive analytics and model deployment.
- **DevOps & CI/CD:** Jenkins, GitHub, UrbanCode Deploy, Kubernetes, Liquibase, Terraform.

PROFESSIONAL EXPERIENCE

AI Systems Engineer | Prompt & Evaluation Architect Oct 2025 – Present

Snorkel AI (Terminus / Hercules / Marlin Project) — Remote

- Lead AI systems and prompt engineering for Snorkel AI's Terminus / Hercules initiative, building deterministic, auditable, failure-resistant LLM evaluation workflows for complex reasoning and system-level tasks.
- Designed constraint-based prompt architectures enforcing single-pass execution, priority semantics, and deterministic outputs, eliminating reliance on retries, backtracking, or hidden chain-of-thought.
- Authored formal task specifications and instruction hierarchies to prevent prompt leakage, shortcut reasoning, and partial-match exploitation in benchmarked AI systems.
- Engineered oracle-grounded evaluation pipelines pairing prompts with reference implementations to validate correctness over fluency for algorithmic and compiler-style tasks.
- Built and validated Dockerized execution environments integrated with CI pipelines for reproducibility, isolation, and runtime compliance.
- Performed LLM failure-mode analysis across reasoning, instruction adherence, state handling, and determinism.
- Collaborated with Snorkel reviewers and platform engineers to ensure benchmarks met enterprise-grade reliability and safety standards.

Senior Data Engineer | Credit Decisioning Transformation Jul 2025 – Oct 2025

Wells Fargo — Dallas, TX

- Led data engineering for the Credit Decisioning Platform, implementing pipelines across ICRDT (Transformation), CODI (Ingestion), and DEPCT (FICO Rules Engine).
- Developed Spark ingestion jobs (T2T table-to-table, F2T file-to-table) with SQL scripts, config files, and Autosys JIL scheduling.
- Built ingestion and transformation pipelines for PCI and HRAM models supporting automated risk-based decisioning; integrated outputs into the FICO Blaze Rules Engine, enabling faster approvals and reducing delinquency exposure.
- Optimized Amazon Redshift performance via query plan analysis, WLM queue tuning, sort keys, and distribution styles; reduced query runtime by 45% on high-volume PCI/HRAM datasets.
- Performed VACUUM/ANALYZE maintenance, compression encoding, and workload scheduling, lowering compute cost by 30%.
- Developed complex SQL stored procedures, CTEs, window functions, and incremental merge strategies for high-throughput decisioning workloads.
- Led benchmarking and SQL tuning across S3 → Glue → Redshift pipelines, improving reliability and SLA consistency.
- Automated deployments with GitHub, Jenkins, and UrbanCode Deploy (DEV/UAT), cutting deployment timelines by 50%; ran unit tests and UAT with business teams.
- Strengthened OCC/CFPB compliance by ensuring ingestion only from certified bureau and internal data sources.

Lead Data Engineer | E-commerce Analytics Project Nov 2023 – Jun 2025

Tagus Technologies LLC — Houston, TX

- Designed and deployed AWS data infrastructure (S3, Glue, EMR, Lambda, Databricks) for e-commerce analytics.
- Built end-to-end ETL pipelines with Python, PySpark, and Apache Airflow, improving reliability by 40%.
- Implemented Databricks Unity Catalog for centralized governance across multiple workspaces, reducing governance overhead by 30%.
- Optimized Delta Lake pipelines (Z-Ordering, clustering, OPTIMIZE/VACUUM), improving query performance by 35% and lowering compute cost.
- Integrated Kafka for real-time streaming, reducing processing latency by 30%.
- Automated Kubernetes-based PySpark deployments with Jenkins and Terraform CI/CD, improving utilization by 35%.
- Tuned Redshift table design (DISTKEY/SORTKEY, compression) and used Spectrum for external tables; reduced ETL runtimes by 40% via SQL refactoring, predicate pushdown, cluster sizing, and partition pruning.
- Built Power BI dashboards (20% operational efficiency gain) and implemented GDPR and SOC2 governance frameworks.
- Built a Microsoft Fabric analytics demo (Lakehouse, Pipelines, Semantic Models, Data Agent, DirectLake) replicating healthcare workloads; automated deployment with Fabric Notebooks and Git-based configuration.
- Developed Synapse SQL pools and Data Factory pipelines for ELT automation.

Data Architect May 2023 – Oct 2023

International Monetary Fund (IMF)

- Designed data models and ETL pipelines for scalability and compliance.
- Built Python validation and profiling scripts, improving dataset reliability.
- Reduced ingestion time by 30% with Airflow and Python workflows.
- Cleared 1,500+ backlog SSIS records, improving SQL Server scheduling.
- Delivered enterprise models with ER/Studio, improving cross-team collaboration.

GCP Senior Data Architect & Developer Dec 2022 – Apr 2023

Verizon — Irving, TX

- Migrated on-prem Hadoop pipelines to Google Cloud (BigQuery, Dataproc, Cloud Storage), boosting performance by 40%.
- Developed Python ETL scripts for automated ingestion, reducing manual work by 50%.
- Optimized Dataproc clusters and BigQuery schemas for telecom analytics.

Software Engineer (Big Data & Java) Oct 2020 – Dec 2021

JPMorgan Chase — Dallas, TX

- Built event-driven microservices with Kafka and Spark Streaming, reducing transaction delays.
- Optimized Spark configurations, improving query execution times by 25%.

- Strengthened compliance with banking data security standards.

Data Engineer – ETL & Analytics Aug 2019 – Oct 2020
Verizon Media — Dallas, TX

- Designed ETL workflows with Python, Databricks, and AWS Glue, improving pipeline efficiency by 30%.
- Created Power BI dashboards for call center analytics, enhancing customer insights.

Big Data Consultant / ETL Developer Apr 2017 – Aug 2019
Deloitte Consulting LLP — Remote / Client Sites

Facebook — Data Engineer

- Designed, built, and launched new production data pipelines and ETL processes using Dataswarm.
- Supported existing production processes; defined and managed SLAs for owned data sets.
- Worked with data infrastructure teams to triage and resolve infrastructure issues.

Kroger — Information Architect

- Analyzed information management requirements for the Floor & Space business domain across all systems and platforms to guide standards aligned with Kroger's enterprise architecture.
- Documented best practices and challenges, and proposed a roadmap to the future state, aligning IT with business objectives.

Fiat Chrysler Automobiles — Big Data Architect

- Engaged stakeholders across 3 regions to understand the as-is state; contributed to architecture and development of a global enterprise big data platform.
- Translated business requirements into system-level technical changes; built models representing business use cases on request.

Dow Jones — Common Data Platform (CDP)

- Moved data from legacy sources to an AWS landing zone; transformed it with Spark and created tables on Amazon Redshift.
- Created Amazon SNS alerts for system errors; managed user stories in Jira and committed code to GitHub.

Anthem Blue Cross — Big Data / ETL Developer

- Moved large volumes of health-insurance/billing data from Teradata into HDFS via the Teradata Hadoop connector; cleansed and transformed with Spark/Scala into Avro and Parquet.
- Configured and monitored a large Cloudera Hadoop cluster (Cloudera Manager), including users/groups, Kerberos, HDFS encryption, and Control-M automation.
- Modernized a legacy COBOL pricing system to Apache Spark (Java and Python) on a 12-node EC2 cluster, processing 100 million records in under 1 minute.

Pioneer Natural Resources — Data Ingestion & Architecture

- Developed on StreamSets, ingesting 8 million records in 15 minutes from Cloudera HDFS and RDBMS sources.
- Delivered a cloud data advisory project with architectural recommendations and investment roadmap, presented to senior stakeholders.

Deloitte Internal Projects

- Designed an Apache Airflow pipeline automating data ingestion to and retrieval from AWS S3 using Python.
- Built a Python AWS Lambda function with S3 trigger moving data from S3 to DynamoDB, with IAM policy and role configuration.

Hadoop Consultant Jun 2015 – Apr 2017
Conquest Consulting LLC — Austin, TX

- Designed Hadoop ETL pipelines with Hive, Pig, Sqoop, and Oozie.
- Built predictive analytics for Dell server logs using HBase, Pig, and Flume.
- Implemented a 15-node Hadoop cluster enabling real-time infrastructure analytics.

RELEVANT PROJECTS

Microsoft Fabric End-to-End Analytics Demo (Healthcare) — Self-Developed, 2025

- Automated installation of Fabric components (Lakehouse, Pipelines, Semantic Model, Power BI, Data Agent) via notebook deployment.

- Engineered ingestion of 275 million healthcare prescription records with efficient partitioning, query performance, and governance.
- Enabled Copilot via Azure OpenAI; built SQL endpoints for drug-cost analytics using DirectLake and Delta tables; designed Power BI dashboards and chat-with-data experiences.

Data Reliability & Incident Response — Wells Fargo, Verizon Media

- Led incident investigations resolving ingestion delays and Spark job failures under strict SLAs.
- Implemented observability with CloudWatch, Grafana, and Spark metrics, reducing Mean Time to Detection by 40%.
- Provided root-cause analysis to business stakeholders and standardized escalation procedures in line with OCC/CFPB mandates.

TECHNICAL SKILLS

- **Cloud & Big Data:** AWS (S3, Glue, Redshift, EMR, Lambda, SageMaker, CloudWatch); Azure (Fabric, Data Factory, Synapse, Data Lake Gen2, Cosmos DB, Logic Apps, Purview, Power BI, Azure OpenAI); GCP (BigQuery, Dataproc, Cloud Composer, Cloud Storage); Databricks (Workflows, Delta Live Tables, Unity Catalog, ML & SQL Warehouses), Delta Lake
- **Data Engineering & ETL:** Apache Spark, PySpark, HiveQL, Informatica, SSIS, AWS Glue, Airflow; Kafka, Spark Streaming, AWS Kinesis; Spark T2T/F2T, Autosys (JIL), UrbanCode Deploy
- **Programming:** Python, SQL (T-SQL, PL/SQL, PostgreSQL, MySQL), Scala, Java; Pandas, NumPy; REST, SOAP, GraphQL
- **Databases:** Oracle, SQL Server, PostgreSQL, IBM DB2, Teradata; MongoDB, DynamoDB, Cassandra, Bigtable
- **Analytics & Modeling:** Power BI, Tableau, Amazon QuickSight; Star Schema, Snowflake, Data Vault
- **DevOps & Infrastructure:** Jenkins, Git/GitHub Actions, Azure DevOps, Kubernetes, Docker, Terraform, Liquibase; Prometheus, Grafana, CloudWatch

EDUCATION

University of California — Analytics Program (2022)
Focus: Data-Driven Decision Making, Predictive Analytics, Business Intelligence
Specialization: Big Data Systems, Distributed Computing, Data Warehousing

CERTIFICATIONS

- AWS Certified Data Analytics – Specialty
- AWS Certified Developer – Associate
- Architecting on AWS (Official AWS Training Certificate)
- Google Cloud Professional Data Engineer
- Microsoft Certified: Azure Data Engineer Associate
- Cloudera Certified Developer for Apache Hadoop (CCDH)
- Cloudera Certified Administrator for Apache Hadoop (CCAH)
- Cloudera University Security Training (On-Demand)